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ARINC 615A Data Loader Using Ethernet Interface for Aircrafts

Anja Veselinović, Branislav M. Todorović, Stanislava Marković, *Member, IEEE*

Abstract — The aim of this paper is to introduce the reader to basic principles of “data loading” protocol along with concepts of Data Loader implementation. The process of firmware update is described as example of usage ARINC 615A Data Loader. ARINC 615A Data Loader allows a user to upload and download data to or from target hardware of an aircraft. ARINC 615A Data Loader provides four operations: FIND, INFORMATION, UPLOAD and DOWNLOAD. All operations are performed by exchanging protocol files over TFTP (Trivial File Transfer Protocol).

Data transfer between Data Loader and LRU (Line Replaceable Unit) in the aircraft network are defined by ARINC 665-3 and ARINC 615A-3 standards.

Keywords — ARINC 615A, ARINC 665-3, ARINC 615A Data Loader, Data Transfer, Trivial File Transfer Protocol

I. INTRODUCTION

The ARINC 615A Data Loader provides uploading or downloading files (Firmware image, configuration, etc.) to or from a network component. ARINC 615A Data Loader is implemented in accordance with standards ARINC 615A-3 [1] and ARINC 665-3 [2].

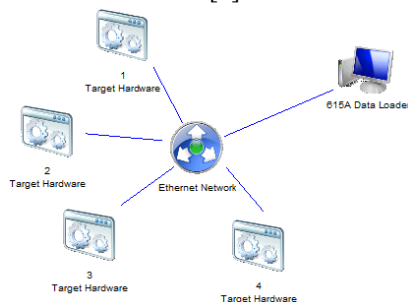


Fig. 1: System Architecture

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A Dataloading Master is a portable unit, remote networking element, Data Loader in ARINC 615A-3. A Dataloading Slave is a Target Hardware in ARINC 615A-3.

ARINC 615A protocol is implemented using the Trivial File Transfer Protocol (TFTP). The ARINC 615A protocol stack is illustrated in Figure 2.

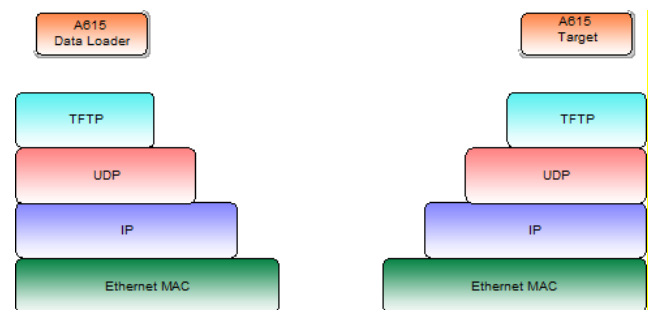


Fig. 2: ARINC 615A Protocol Stack

ARINC 615A-3 standard defines that all operations as well as the sending of status files shall use TFTP.

The purpose of this paper is to present general design guidance for the development of Data Loader for all types of aircraft. Section II gives an overview of ARINC standards which define a "data loading" protocol. Section III describes a protocol that handles the transfer of files. Data loader operations and error handling are presented in sections IV and V. An example of firmware update using ARINC 615 Data Loader is presented in section VI.

II. ARINC STANDARD OVERVIEW

ARINC publishes standards produced by the Airlines Electronic Engineering Committee (AEEC). The AEEC is an international standards organization made up of major airline operators, avionics industry manufacturers and ARINC members.

ARINC 615A defines the standard file and data transfer method for transferring data and software files between Portable Data Loader (PDL) and Loadable Target Hardware – LRU (Line Replaceable Unit). The Data Loader communicates with the Loadable Hardware via an Ethernet Interface. Internet Protocol (IP), User Datagram Protocol (UDP) and Trivial File Transfer Protocol (TFTP) are used by ARINC 615A to provide network and file transfer services.

ARINC 665-3 standard defines standards for loadable software parts and software transport media. It describes

the principles and rules to be applied to any part of a data load system, to ensure compatibility. It includes part numbering, content, labeling and formatting of loadable software parts and software transport media.

III. TRIVIAL FILE TRANSFER PROTOCOL (TFTP)

The TFTP, defined in RFC 1350 [3], is a protocol that handles the transfer of files. TFTP is implemented on top of the UDP (User Datagram Protocol) protocol. A TFTP transfer begins with the TFTP client sending a read or write request to a TFTP server on a well-known port (69 is the dedicated port for TFTP, 59 is the ARINC 615A-3 dedicated port). If the TFTP server accepts the request, a connection is established for the duration of the file transfer and closed afterwards. The protocol only allows a client to read and write files from or to a remote server. [3]

TFTP supports five types of packets: Read Request (RRQ), Write Request (WRQ), Data (DATA), Acknowledgement (ACK) and Error (ERROR). How these packets interact to allow a client to upload and download files to TFTP server is depicted in Figure 3 and Figure 4. [4]

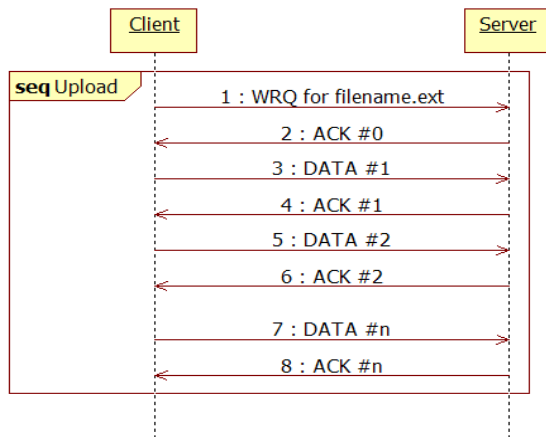


Fig. 3: TFTP upload

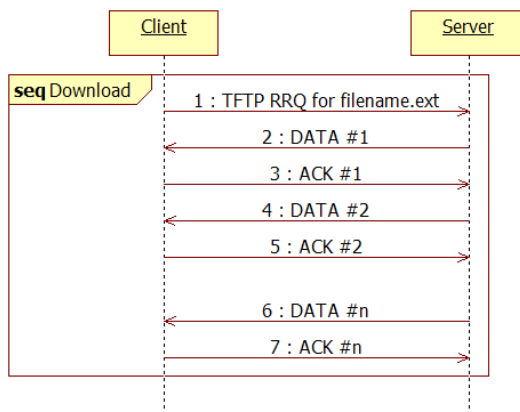


Fig. 4: TFTP download

Read Request and Write request: To establish a transfer the client sends a Read Request (RRQ) or a Write Request (WRQ) to the server. The server responds by sending the Data packet (DATA) if the request was PRQ, or an Acknowledgment (ACK) packet if the request was WRQ.

Data: The Data packet contains the actual data of the file to be transferred with a block number to identify it. After Data packet is sent, the sender waits for acknowledgement before sending the next segment of the data.

Acknowledgement: Upon receiving a Data packet, the server sends an ACK packet. This packet is also used by server to grant a WRQ.

The ARINC 615A Data loader requires two instances of the TFTP service. One instance is needed by the main operation task (handling the steps of operation) and the other instance is needed by the status task (sending status frames periodically).

IV. ARINC 615A DATA LOADER OPERATIONS

The ARINC 615A Data Loader supports four operations according to the ARINC 615A-3 Standard:

- FIND
- INFORMATION
- UPLOAD
- DOWNLOAD

The Upload operation transfers files from the Dataloading Master (Data Loader) to the Dataloading Slave (Target Hardware). The Download operation transfers files from the Dataloading Slave (Target Hardware) to the Dataloading Master (Data Loader).

User files (data files) are transferred via TFTP and ARINC 615A protocol information is also transferred as file (protocol file).

A. FIND Operation

FIND operation is used to determine which Target Hardware is present on the network and to obtain minimal information to establish communications. The ARINC 615A Data Loader responds to FIND request received on port 1001 with a FIND answer.

If the FIND operation is not accepted, the ARINC 615A Data Loader shall ignore the FIND request.

This operation is useful in identifying correct hardware on which load operation has to be performed.

B. INFORMATION Operation

Information operation is started if these two conditions are satisfied:

- TFTP request for Load Configuration Initialization (LCI) file is received and
- Information operation is accepted.

If the TFTP layer requests reading of Load Configuration Initialization (LCI) file, the ARINC 615A Data Loader shall provide a LCI file with content defined in accordance with the ARINC 615A-3 [1] standard. After information operation has started, the ARINC 615A Data Loader shall send the Load Configuration List (LCL) file, which contains Target Hardware information, to the Dataloading master. Status task shall start sending Load Configuration Status (LCS) files reporting that operation is in progress.

LCL file contains software part numbers and version information for the loadable software parts. The format of this information is defined in ARINC 615A standard within the definition of the LCL File. Figure 5 presents the content of LCL file.

Hardware Code	ETH_TARGET1
Serial Number	PN-12345-67
Loadable Parts	
Part Number	SWPN-1234
Amendment	Rev. A
Part Designation	BIU1-FW
Part Number	SWPN-5678
Amendment	Rev. B
Part Designation	BIU2-FW
Hardware Code	ETH_TARGET2
Serial Number	PN-12345-89
Loadable Parts	
Part Number	SWPN-8901
Amendment	Rev. C
Part Designation	IO-LCA SRE

Fig. 5: Content of LCL file

C. UPLOAD Operation

This operation helps in uploading new or updated software, data and configuration files to the Target Hardware. According to ARINC 615A-3 standard the ARINC 615A Data Loader shall start the upload operation if:

- a TFTP request for Load Uploading Initialization (LUI) file has been received and
- upload operation has been accepted.

If the TFTP layer requests reading of Load Uploading Initialization (LUI) file, the 615A Data Loader shall provide a LUI file with content as defined in the ARINC 615A-3 [1] specification. This file tells the operator whether the Target Hardware supports the operation, is ready for it and what version of ARINC 615A it runs. After upload operation has started, the 615A Status Task shall start sending Load Uploading Status (LUS) files reporting "operation in progress". The ARINC 615A Data Loader shall accept only Load Upload Request (LUR) files with number of header files greater than zero and lesser or equal to the configured maximal number of header files. Zero header files means that there is nothing to upload. Allowing more files than defined maximum would compromise the boundaries defined by the integrator and could result in a not defined behavior.

During upload operation, after LUR file was received and accepted, the 615A Data Loader shall request a transfer of all header files that were listed in LUR file. After all header files were received and accepted, the 615A-3 Data Loader shall request a transfer of all data files and all support files that were listed in each header file.

The status file gives the Data Loader continuous updates on how the operation is progressing from the Target Hardware's perspective. It also serves as a "heartbeat" that allows the Target Hardware to indicate that it is working

even though it might not currently request files. The ARINC 615A Data Loader shall:

- calculate a 16bit CRC (cyclic redundancy check) of a received Load Upload Header (LUH) file,
- verify the calculated CRC value against the CRC value stored in LUH file
- indicate the verification result via the check file.

Calculation and the polynomial for CRC are defined in ARINC REPORT 665-3 [2]. Figure 6 shows flow diagram of Upload Operation.

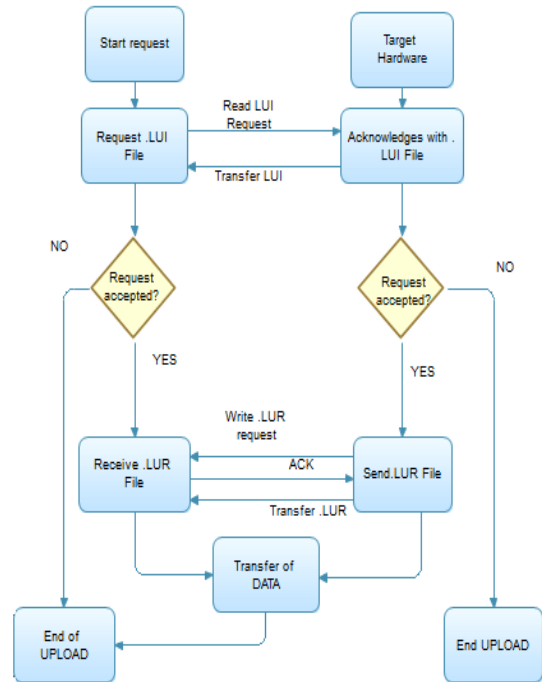


Fig. 6: Flow diagram of UPLOAD Operation

D. DOWNLOAD Operation

This operation allows the operator to download data files from the Target Hardware.

The 615A-3 Data Loader shall start the download operation if:

- a TFTP request for Load Downloading Operator (LNO) file has been received and
- download operation has been accepted

Figure 7 shows flow diagram of Download Operation.

The 615A Data Loader shall provide a LNO file with content as defined in the ARINC 615A-3 [1] specification. After download operation has started, the 615A-3 Status Task shall start sending Load Downloading Status (LNS) files reporting "operation in progress".

If the TFTP layer requests reading of Load Downloading Status file, the 615A-3 Data Loader shall provide a LNS file with content as defined in the ARINC 615A-3 [6] specification.

After download operation has started, the 615A-3 Data Loader shall send the Load Downloading List (LNL) file to the Dataloading master. If the TFTP layer requests reading of Load Downloading List (LNL) file, the 615A Data Loader shall provide a LNL file with content as defined in the ARINC 615A-3 [1] specification. During download operation, after Load Downloading Answer

(LNA) file was received, the 615A-3 Data Loader shall request a transfer of all data files that were listed in the LNA file.

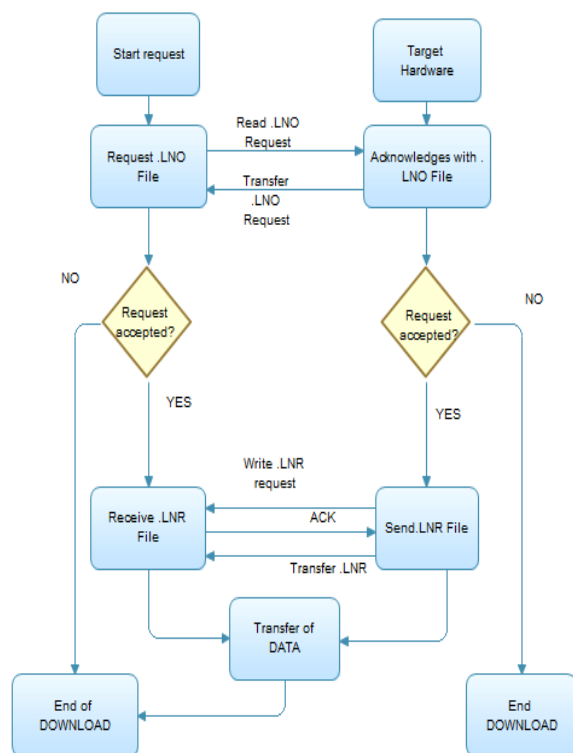


Fig. 7: Flow diagram of DOWNLOAD Operation

V. ERROR HANDLING

If an error occurs during the transfer of a file, an error packet is sent to indicate the type of error. This packet is followed by an error code value. ARINC 615A defines two types of errors:

WAIT Message: The WAIT Message can be sent in response to an initial request (WRQ or RRQ). This message is used to tell the client to wait before trying again. The period of wait time is specified in the wait message. The format of this TFTP error with a wait time of 50 seconds is presented in Figure 8.

OpCode	Error Code Value	Error Message (text)	ASCII String Termination
0x0005	0x0000	“WAIT:50”	0x00

Fig. 8: WAIT Message

ABORT Message: The ABORT Message is used to tell the client to abort operation currently in process. If a TFTP error is caused by timeout, the ARINC 615A Data Loader shall abort the current operation only if the maximal number of retries was reached. The format of this error is shown in Figure 9.

OpCode	Error Code Value	Error Message (text)	ASCII String Termination
0x0005	0x0000	“ABORT:XXXX”	0x00

Fig. 9: ABORT Message

The “XXXX” string corresponds to hexadecimal value of the Data Load Status Code, defined in ARINC 615A-3 standard.

VI. EXAMPLE OF FIRMWARE UPDATE USING ARINC 615A DATA LOADER

One of the possibilities that ARINC 615A Data Loader provides is firmware image update.

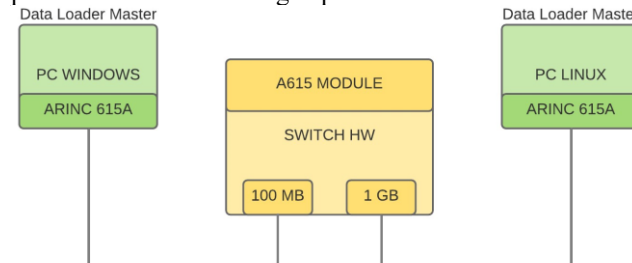


Fig. 10: Data loading architecture

ARINC A615A Data Loader Master (DLM) is used to upload firmware file to switch hardware with integrated ARINC 615A module (Target hardware), presented in Figure 10. The switch core software controls the behaviour of ARINC 615A dataloading through the configuration and through the callbacks. If a new version of firmware is uploaded, in callback “ARINC 615A operation ends” firmware activation is triggered. After switch reset activation will be started.

In this example standard Ethernet was used for data loading between Data Loader Master (on Windows PC as well as on Linux PC) and Target Hardware as network interface. ARINC 615A protocol also provides data loading using Avionics Full-Duplex Switched Ethernet (AFDX) which is based on ARINC 664 standard and it is full duplex, deterministic and high speed network.

VII. CONCLUSION

Avionics industries continue to become more focused on software centric avionics systems and it has become important to have standard methods of loading software and data into the avionics systems. ARINC 615A Data Loader provides possibility that multiple users can operate Data Loader simultaneously without care about manufacturer of the Target Hardware which is used.

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